MODEL: 6060 ● Bucket Size: 44.5 yd³ ● Number of Cycles: 4 ● Material Density: 3030 lb/yd³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		67.5 tons				64.0 tons			
	Load per Truck		270 tons				256 tons			
Overall Mine Efficiency	Cycle Time		26 sec	28 sec	30 sec	32 sec	26 sec	28 sec	30 sec	32 sec
	Truck Exchange		0.43 min	0.47 min	0.50 min	0.53 min	0.43 min	0.47 min	0.50 min	0.53 min
83% = maximum 50 min/hour	30 sec	0.50 min	7303 tons/h	6856 tons/h	6555 tons/h	6280 tons/h	6933 tons/h	6509 tons/h	6223 tons/h	5962 tons/h
	42 sec	0.70 min	6587 tons/h	6221 tons/h	5973 tons/h	5743 tons/h	6254 tons/h	5906 tons/h	5670 tons/h	5452 tons/h
	54 sec	0.90 min	5999 tons/h	5694 tons/h	5485 tons/h	5291 tons/h	5695 tons/h	5406 tons/h	5207 tons/h	5023 tons/h
	66 sec	1.10 min	5508 tons/h	5249 tons/h	5071 tons/h	4904 tons/h	5229 tons/h	4983 tons/h	4814 tons/h	4656 tons/h
80%	30 sec	0.50 min	7013 tons/h	6584 tons/h	6295 tons/h	6030 tons/h	6658 tons/h	6250 tons/h	5976 tons/h	5724 tons/h
	42 sec	0.70 min	6326 tons/h	5974 tons/h	5735 tons/h	5515 tons/h	6005 tons/h	5671 tons/h	5445 tons/h	5235 tons/h
	54 sec	0.90 min	5761 tons/h	5468 tons/h	5267 tons/h	5080 tons/h	5469 tons/h	5191 tons/h	5000 tons/h	4823 tons/h
	66 sec	1.10 min	5289 tons/h	5041 tons/h	4869 tons/h	4709 tons/h	5021 tons/h	4785 tons/h	4623 tons/h	4471 tons/h
75%	30 sec	0.50 min	6573 tons/h	6171 tons/h	5900 tons/h	5652 tons/h	6240 tons/h	5858 tons/h	5601 tons/h	5365 tons/h
	42 sec	0.70 min	5929 tons/h	5599 tons/h	5375 tons/h	5169 tons/h	5628 tons/h	5316 tons/h	5103 tons/h	4907 tons/h
	54 sec	0.90 min	5399 tons/h	5125 tons/h	4937 tons/h	4762 tons/h	5126 tons/h	4865 tons/h	4686 tons/h	4520 tons/h
	66 sec	1.10 min	4957 tons/h	4724 tons/h	4564 tons/h	4414 tons/h	4706 tons/h	4485 tons/h	4333 tons/h	4190 tons/h
70%	30 sec	0.50 min	6133 tons/h	5758 tons/h	5505 tons/h	5273 tons/h	5822 tons/h	5466 tons/h	5226 tons/h	5006 tons/h
	42 sec	0.70 min	5532 tons/h	5225 tons/h	5016 tons/h	4823 tons/h	5252 tons/h	4960 tons/h	4761 tons/h	4578 tons/h
	54 sec	0.90 min	5038 tons/h	4782 tons/h	4606 tons/h	4443 tons/h	4783 tons/h	4540 tons/h	4373 tons/h	4218 tons/h
	66 sec	1.10 min	4625 tons/h	4408 tons/h	4258 tons/h	4119 tons/h	4391 tons/h	4185 tons/h	4043 tons/h	3910 tons/h
65%	30 sec	0.50 min	5693 tons/h	5345 tons/h	5110 tons/h	4895 tons/h	5405 tons/h	5074 tons/h	4851 tons/h	4647 tons/h
	42 sec	0.70 min	5135 tons/h	4850 tons/h	4656 tons/h	4477 tons/h	4875 tons/h	4604 tons/h	4420 tons/h	4250 tons/h
	54 sec	0.90 min	4677 tons/h	4439 tons/h	4276 tons/h	4124 tons/h	4440 tons/h	4214 tons/h	4059 tons/h	3915 tons/h
	66 sec	1.10 min	4293 tons/h	4092 tons/h	3953 tons/h	3823 tons/h	4076 tons/h	3885 tons/h	3753 tons/h	3629 tons/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		60.7 tons				57.3 tons			
	Load per Truck		243 tons				229 tons			
Overall Mine Efficiency	Cycle Time		26 sec	28 sec	30 sec	32 sec	26 sec	28 sec	30 sec	32 sec
	Truck Exchange		0.43 min	0.47 min	0.50 min	0.53 min	0.43 min	0.47 min	0.50 min	0.53 min
83% = maximum 50 min/hour	30 sec	0.50 min	6575 tons/h	6173 tons/h	5902 tons/h	5654 tons/h	6206 tons/h	5826 tons/h	5570 tons/h	5336 tons/h
	42 sec	0.70 min	5931 tons/h	5601 tons/h	5377 tons/h	5170 tons/h	5597 tons/h	5286 tons/h	5075 tons/h	4880 tons/h
	54 sec	0.90 min	5401 tons/h	5127 tons/h	4938 tons/h	4763 tons/h	5097 tons/h	4838 tons/h	4660 tons/h	4495 tons/h
	66 sec	1.10 min	4959 tons/h	4726 tons/h	4566 tons/h	4416 tons/h	4680 tons/h	4460 tons/h	4309 tons/h	4167 tons/h
80%	30 sec	0.50 min	6314 tons/h	5927 tons/h	5667 tons/h	5429 tons/h	5959 tons/h	5594 tons/h	5348 tons/h	5123 tons/h
	42 sec	0.70 min	5695 tons/h	5379 tons/h	5163 tons/h	4965 tons/h	5375 tons/h	5076 tons/h	4873 tons/h	4686 tons/h
	54 sec	0.90 min	5187 tons/h	4923 tons/h	4742 tons/h	4574 tons/h	4895 tons/h	4646 tons/h	4475 tons/h	4317 tons/h
	66 sec	1.10 min	4761 tons/h	4538 tons/h	4384 tons/h	4240 tons/h	4494 tons/h	4283 tons/h	4137 tons/h	4002 tons/h
75%	30 sec	0.50 min	5918 tons/h	5556 tons/h	5312 tons/h	5088 tons/h	5585 tons/h	5243 tons/h	5013 tons/h	4802 tons/h
	42 sec	0.70 min	5338 tons/h	5041 tons/h	4840 tons/h	4653 tons/h	5037 tons/h	4758 tons/h	4567 tons/h	4392 tons/h
	54 sec	0.90 min	4861 tons/h	4614 tons/h	4444 tons/h	4287 tons/h	4588 tons/h	4354 tons/h	4194 tons/h	4046 tons/h
	66 sec	1.10 min	4463 tons/h	4254 tons/h	4109 tons/h	3974 tons/h	4212 tons/h	4014 tons/h	3878 tons/h	3750 tons/h
70%	30 sec	0.50 min	5522 tons/h	5184 tons/h	4956 tons/h	4748 tons/h	5211 tons/h	4892 tons/h	4677 tons/h	4481 tons/h
	42 sec	0.70 min	4980 tons/h	4704 tons/h	4516 tons/h	4342 tons/h	4700 tons/h	4439 tons/h	4262 tons/h	4098 tons/h
	54 sec	0.90 min	4536 tons/h	4305 tons/h	4147 tons/h	4000 tons/h	4281 tons/h	4063 tons/h	3914 tons/h	3775 tons/h
	66 sec	1.10 min	4164 tons/h	3969 tons/h	3834 tons/h	3708 tons/h	3930 tons/h	3746 tons/h	3618 tons/h	3499 tons/h
65%	30 sec	0.50 min	5126 tons/h	4812 tons/h	4601 tons/h	4407 tons/h	4837 tons/h	4541 tons/h	4342 tons/h	4159 tons/h
	42 sec	0.70 min	4623 tons/h	4366 tons/h	4192 tons/h	4030 tons/h	4363 tons/h	4121 tons/h	3956 tons/h	3804 tons/h
	54 sec	0.90 min	4210 tons/h	3996 tons/h	3850 tons/h	3713 tons/h	3974 tons/h	3771 tons/h	3633 tons/h	3504 tons/h
	66 sec	1.10 min	3865 tons/h	3684 tons/h	3559 tons/h	3442 tons/h	3648 tons/h	3477 tons/h	3359 tons/h	3248 tons/h

TELESCOPIC HANDLERS

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Features 14-1

Features:

- **Intuitive control layout** enables efficient and comfortable operation.
- **Hydraulic system** enables the use of work tools requiring both intermittent and continuous auxiliary hydraulic flows.
- **Range of lift heights and load capacities** from 2500 kg (5,500 lb) to 5443 kg (12,000 lb) capacity and 5.6 m (18'4") to 17.3 m (56'9") lift height.
- **Cat® Telehandlers offer extended reach** to easily enable material to be delivered to high elevations or placed over onsite obstacles.

Features *(continued)*:

- **Three steering modes and tight turning circles** allow simplified movement onsite and therefore increased productivity.
- **Cat Telehandlers have engine options** which either meet U.S. EPA Tier 4 Final emission standards, EU Stage V emission standards, or U.S. EPA Tier 3/ Stage IIIA equivalent emission standards.
- **Three transmissions options are available on telehandlers.** The TH255 is fitted with a hydrostatic transmission. The TL and TH construction range are fitted with 4-speed powershift transmissions. The high horsepower TH Ag Handler range is fitted with a 6-speed powershift transmission.
- **Three types of quick couplers are offered on the various telehandler models.** The TH255C is fitted with a skid-steer loader style universal coupler. The TL and TH lines of telehandlers offer hydraulic and manual style IT interface couplers which provide excellent work tool compatibility with older models of telehandlers and other products (for example Backhoe Loaders and Small Wheel Loaders).
- **Hydraulic Options** – On all product models single auxiliary service is standard and Hydraulic quick coupler is an option (not considered a separate hydraulic service). For the TH357, TH408, and TH3510 models, Continuous Flow is a standard feature, with Dual or Secondary Auxiliary Hydraulic service as an option. These options allow maximum work tool utilization.
- **Daily maintenance** consists of only a walkaround inspection, all major components and service points are located at ground level, providing excellent ease of service.

TRACK LOADERS

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Features of 953-963:

One Machine, Many Jobs

- Track loader power, traction and agility mean you can use one machine and one operator to do everything from clearing the site to installing utilities and finishing around buildings.
- Fusion™ Quick Coupler option lets you make fast attachment changes with a wide range of tools like forks, buckets, etc. from wheel loaders and other Fusion compatible machines.
- Bucket/tooth, undercarriage, track shoe and attachment choices help you equip your machine for optimal performance in your applications.
- Purpose-built Waste Handler, Ship Hold and Low Ground Pressure (LGP) configurations stand up to the toughest tasks.

Easy to Operate

- Updated cab offers more comfort and an intuitive 254 mm (10-inch) touchscreen dash display.
- Standard High Definition rearview camera shows prominently in the main display.
- Slope Indicate helps make operation easier by showing machine mainfall and cross slope right on the display.
- Enjoy operator comforts like suspension seat and adjustable armrests and controls.
- Choose either joystick or two-lever implement controls to match operator preference or application.
- Speed/steering control choices include either a joystick or a v-lever and foot pedals.
- Set implement response – fine, normal, coarse – to match operator preference or application.
- Emissions reduction technology is designed to work in the background, with no action required from the operator.

Top Performer

- 963 and 953 uses up to 10% less fuel per hour than the respective previous model.
- Performance Series bucket option boosts productivity up to 20% compared to General Purpose bucket.
- Efficient Cat C7.1 engine delivers power and fast response when the machine is under load.
- Meets U.S. EPA Tier 4 Final/EU Stage V/Japan 2014/ Korea Tier 4 Final emission standards.
- Smoother implement and steering response, and improved steering performance provide more controllability.
- Load sensing hydraulics and position sensing cylinders offer precise, responsive performance.

Save on Maintenance

- Designed to help you take care of routine maintenance and get back to work.
- Grouped service points located behind large access doors; daily grease points in easy reach at ground level.
- Service mode in the operator display shows hydrostatic and implement pressure for easier troubleshooting and servicing.
- Demand fan provides efficient cooling plus easy fold-down service access.
- Handy bracket holds a shovel for quick undercarriage clean-out.
- Tilt cab and convenient service access points for easier maintenance and repair.
- Diesel Particulate Filter in the Clean Emissions Module designed to work for the life of the engine without needing to clean or replace the filter.

Features of 953-963 ***(continued):***

Focus on Safety

- Excellent visibility to the bucket and all around the machine enhances safety and helps operators work more confidently.
- Handles and steps help operators climb on and off the machine more easily, from the front or the back of the tracks.
- Standard rear vision camera enhances visibility behind the machine.
- Seat belt indicator gives an alarm and registers fault code through Product Link™ if the operator fails to buckle up, enhancing job site safety.
- Ergonomics, a quiet engine and low-effort controls help lessen fatigue, so operators are better able to remain fresh and focused.

Take on the Trash

- Waste Handler configuration is a versatile machine for loading, sorting, excavation and spreading cover, well suited to the landfill or transfer station.
- Specialized guarding, striker bars and seals help protect the machine and components from impact and airborne debris.
- Final Drive guarding helps prevent wrapping and damage.
- Screen helps protect windshield and operator from breakage and debris.
- High debris cooling system – radiator fan folds out for easy cleanout access.
- Specialized air handling features help deliver cleaner air to the machine and to the cab.
- Choice of buckets and track shoes allow you to further optimize the machine for greater performance and service life.
- Enhanced Cleaning Package reduces cleaning time, adds machine protection and increases compaction.

Keep Things Shipshape

- Optimal combination of traction, high reach and machine balance makes track loaders ideal for working in ship holds and port handling duties.
- Specialized implements, like coal buckets and trim blades, help you sweep down walls and handle a variety of materials.
- Fusion Quick Coupler option adds versatility with easy use of forks, buckets and other attachments from compatible machines.
- Specialized sealing, guards and bumpers help protect key components.
- Front and rear eyes widely spaced for stability during lifting.
- Added lighting packages help illuminate the work area.

EXTREME SLOPE OPERATION

The maximum fore and aft grade in static condition on which each track loader will maintain proper lubrication is 45 degrees (100%). Consult Operation & Maintenance Manual (if applicable) for POWERTRAIN fluid level overfill requirements for operation on extreme slopes. Extreme slope operation is anytime the slope exceeds 25° (47%).

The ENGINE should never be overfilled with oil. This may lead to rapid overheating. For extreme slope operation, engine oil should be maintained at the full mark.

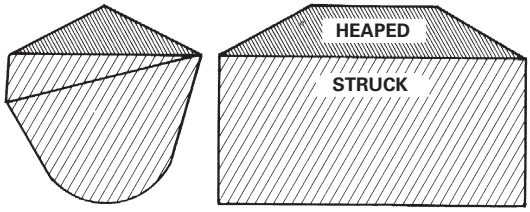
NOTE: Both ENGINE and POWERTRAIN fluid levels should be checked on level ground before working sidehills and slopes.

When working sidehills and slopes, consideration should be given to the following important points:

- Speed of travel — At higher speeds, inertia forces tend to make the track loader less stable.
- Roughness of terrain or surface — Ample allowance should be made where the terrain or surface is uneven.
- Mounted equipment — Buckets, rippers, winches, and other mounted equipment cause the track loader to balance differently.
- Nature of surface — New earthen fills may give way with the weight of the track loader. Rocky surfaces may promote side slipping of track loader.
- Track slippage due to excessive loads — This may cause downhill track to "dig in," increasing angle of track loader.
- Width of shoes — Wide track shoes tend to decrease "digging in," hence track loader is more stable.
- Operated equipment — Be aware of the stability and other performance features of the equipment operated by the track loader.
- Keep all attachments or pulled loads low to the ground for optimum stability.

NOTE: Safe operation on steep slopes may require special machine maintenance as well as excellent operator skill and proper equipment for the specific application. Consult Operation & Maintenance Manual (if applicable) for proper fluid level requirements.

SAE BUCKET RATING



SAE Bucket Capacities

Struck capacity is that volume contained in a bucket after a load is leveled by drawing a straight edge resting on the cutting edge and the back of the bucket.

Heaped capacity is a struck capacity *plus* that additional material that would heap on the struck load at a 2:1 angle of repose with the struck line parallel to the ground.

SAE J742 (Oct. 79) specifies that the addition of any auxiliary spill guard to protect against spillage of material which might injure the operator will not be included in bucket capacity ratings. Buckets with irregular shaped cutting edges (vee edge) the strike plane should be drawn at one-third the distance of the protruding portion of the cutting edge. Cat rock buckets are built with integral see-through rock guards. Cat light material buckets come standard with bolt-on edges. These features which add to actual bucket capacity are included in published ratings.

Dump Height

SAE J732 JUN92 specifies that dump height is the vertical distance from the ground to the lowest point of the cutting edge with the bucket hinge pin at maximum height and the bucket at a 45° dump angle. Dump angle is the angle in degrees that the longest flat section of the inside bottom of the bucket will rotate below horizontal.

Static Tipping Load

The minimum weight at center of gravity of "SAE Rated" load in bucket which will rotate rear of machine to a point where, on track loaders, front rollers are clear of the track under the following conditions:

a. Loader on hard level surface and stationary.
b. Unit at standard operating weight.
c. Bucket at maximum rollback position.
d. Load at maximum forward position during raising cycle.
e. Unit with standard equipment as described in specifications unless otherwise noted under the heading.

Operating Load

In order to comply with SAE standard J818 MAY87, the operating load for track loaders should not exceed 35% of the Static Tipping load rating. See "Performance Data" of each machine in this handbook for increases to static tipping load by adding cab, counterweights, ripper-scarifier, etc.

SELECTING A MACHINE

Steps in selecting the proper size loader:

1. Determine production required or desired.
2. Determine loader cycle time and cycles per hour. A machine size must be assumed to select a basic cycle time.
3. Determine required payload per cycle in loose cubic yards and pounds (meters and kilograms).
4. Determine bucket size needed.
5. Make machine selection using bucket size and payload as criteria to meet production requirements.
6. Compare the loader cycle time used in calculations to the cycle time of the machine selected. If there is a difference, rework the process beginning at step 2.

1. Production Required

The production required of a track loader should be slightly greater than the production capability of the other critical units in the earth or material moving system. For example, if a hopper can handle 300 tons per hour, a loader capable of slightly more than 300 tons should be used. Required production should be carefully calculated so the proper machine and bucket selections are made.

2. Loader Cycle Times

Material type, pile height, and other factors may improve or reduce production, and should be added to or subtracted from the basic cycle time when applicable.

When hauls are involved, obtain haul and return portions of the cycle from the estimated travel chart (this section). Add the haul and return times to the estimated basic cycle time to obtain total cycle time.

CYCLE TIME FACTORS

A basic cycle time (Load, Dump, Maneuver) of 0.25-0.35 minutes is average for a track loader [the basic cycle for large track loaders, 2 m^3 (2.6 yd^3) and up, can be slightly longer], but variations can be authenticated in the field. The following values for many variable elements are based on normal operations. Adding or subtracting any of the variable times will give the total basic cycle time.

Estimating Cycle Time

Cycle time of a track loader needs to be determined to find loads per hour. Total cycle time includes the following segments:

Load Time + Maneuver Time + Travel Time + Dump Time

Load Time —

Material	Minutes
Uniform aggregates	0.03-0.05
Moist mixed aggregates	0.03-0.06
Moist loam	0.03-0.07
Soil, boulders, roots	0.04-0.20
Cemented materials	0.05-0.20

Maneuver Time — includes basic travel, four changes of direction and turning time, and will be about 0.20 minutes with a competent operator.

Travel Time — in a load and carry operation is comprised of haul and return times which can be determined by the travel charts in this section.

Dump Time — is dictated by the size and strength of the dump target and varies from 0.00 to 0.10 minutes. Typical dump times into highway trucks are from 0.04 to 0.07 minutes.

NOTE: When comparing hydrostatic track loaders with former power shift models (using the production estimating method) two factors must be considered: (1) The hydrostatic track loaders on the average outcycle power shift models by up to 10 percent due to faster machine speed and easier operation. (2) Larger, rear engine hydrostatic track loaders incorporate Z-bar linkage, which provides substantially better bucket fill factors. The degree to which each factor affects estimated production should be left to the user's judgment depending on the particular job application and conditions.

Example: Moist loam is being excavated from a bank and loaded into trucks.

	Minutes
Load — moist loam	0.05
Maneuver Time	0.20
Travel — none required	0.00
Dump	0.05
Total Cycle	0.30 min. or 200 cycles per 60 min. hour

	Minutes added (+) or Subtracted (–) From Basic Cycle
Materials	
— Mixed	+0.02
— Up to 3 mm (1/8 in)	+0.02
— 3 mm (1/8 in) to 20 mm (3/4 in)	–0.02
— 20 mm (3/4 in) to 150 mm (6 in)	0.00
— 150 mm (6 in) and over	+0.03 and Up
— Bank or broken	+0.04 and Up
Pile	
— Conveyor or Dozer piled 3 m (10 ft) and up	0.00
— Conveyor or Dozer piled 3 m (10 ft) or less	+0.01
— Dumped by truck	+0.02
Miscellaneous	
— Common ownership of trucks and loaders	Up to –0.04
— Independently owned trucks	Up to +0.04
— Constant operation	Up to –0.04
— Inconsistent operation	Up to +0.04
— Small target	Up to +0.04
— Fragile target	Up to +0.05

Using actual job conditions and the above factors, total cycle time can be estimated. Convert total cycle time to cycles per hour.

$$\textit{Cycles per hour at 100\% Efficiency} = \frac{60\ \textit{Min}}{\textit{Total Cycle Time in Minutes}}$$

Job efficiency is an important factor in machine selection. Efficiency is the actual number of minutes worked during an hour. Job efficiency accounts for operator breaks, and other work interruptions. See "Efficiency Considerations" in this section.

Bucket Fill Factors

The following indicates the approximate amounts of material as a percent of rated bucket capacity which will actually be delivered per bucket per cycle. This is known as "Bucket Fill Factor."

Loose Material	Fill Factor
Mixed Moist Aggregates	95-110%
Uniform Aggregates	
up to 3 mm (1/8 in)	95-110
3 mm-9 mm (1/8 in-3/8 in)	90-110
12 mm-20 mm (1/2 in-3/4 in)	90-110
24 mm and over (1 in)	90-110
Blasted Rock	
Well	80-95%
Average	75-90
Poor	60-75
Other	
Rock Dirt Mixtures	100-120%
Moist Loam	100-120
Soil, Boulders, Roots	80-100
Cemented Materials	85-100

Fill factors on track loaders are affected by bucket penetration, breakout force, rack back angle, bucket profile and ground engaging tools such as bucket teeth and segments or bolt-on replaceable cutting edges.

GENERAL PURPOSE BUCKET W/TEETH & SEGMENTS MAXIMUM OPERATING CAPACITIES

	GENERAL PURPOSE BUCKET SIZE		MAXIMUM OPERATING CAPACITY	
MODEL	m^3	yd^3	kg	lb
953D/953K	1.85	**2.4**	3182	**7015**
963D/963K	2.45	**3.2**	4214	**9290**
973D	3.21	**4.2**	5521	**12,174**

LOADER PRODUCTION

Loader production equals quantity of material the bucket carries per load × number of bucket loads per hour.

Estimating Bucket Load

The quantity of material in a loader bucket is estimated by two methods, depending on whether the material being loaded is in a loose or bank state.

1. When the material is loose, as in stockpile loading, the bucket load is estimated in loose meters (or cubic yards) by a Bucket Fill Factor (see Tables Section or chart following this discussion). The quantity of material is determined as follows:
   Rated Bucket Capacity × Bucket Fill Factor = Bucket Payload in Loose m^3 (yd^3)

   For example, a 973 with a 3.2 m^3 (4.2 yd^3) General Purpose bucket loading moist loam material will carry:
   3.2 m^3 × 1.15 = 3.68 loose cubic meters
   (4.2 yd^3 × 1.15 = 4.83 loose cubic yards)

   Once the potential bucket load has been determined, check the static tipping load ratings on the specific machine to determine if bucket load is in fact a safe operating load. *(Safe operating load as defined by SAE for track loaders should not exceed 35% of static tipping load.)*

   Productivity in many applications is measured in tons. See Tables Section for material densities if conversion to tons is desired.

2. When material is in the bank state, as in excavation, productivity is measured in bank meters (cubic yards). Bucket load in Bm^3 (BCY) is estimated by applying one of the load factors from the Tables section to convert the excavated material in the bucket from Bm^3 (BCY) to Lm^3 (LCY) to allow for the digging and carrying characteristics of the material. The quantity of excavated material a bucket carries is then determined as follows:
   Rated Bucket Capacity × Load Factor × Bucket Fill Factor = Bucket Payload in Bm^3 (BCY)

   Example: a 953D with a 1.85 m^3 (2.4 yd^3) General Purpose bucket loading wet loam earth from bank:
   1.85 m^3 × 0.79 × 1.15 = 1.68 Bm^3
   (2.4 yd^3 × 0.79 × 1.15 = 2.18 BCY)

Estimating Production

Machine and job considerations include:
- Machine model and bucket size
- Material type, particle size, density and load factor (see Tables Section)
- Bucket fill factor
- Haul distance
- Underfoot conditions
- Altitude
- Dump target size, height, and type

Example:

Conditions —

Machine	953D
Bucket size	1.85 m^3 (2.4 yd^3)
Material	Moist Loam
Bucket fill factor	1.15
Haul length	30 m (100 ft)
Dump target	Pile
Travel in forward speed	

Cycle Time	**Minutes**
Load time	0.15
Maneuver time	0.20
Travel time (from curves)	0.40
Dump time	0.05
Total	0.80

Loads Per Hour —

$$\frac{60 \text{ min/hr}}{0.80 \text{ min/cycle}} = 75 \text{ cycles per hour @ } 100\% \text{ efficiency}$$

Load Per Cycle —

1.85 m^3 × 1.15 BFF = 2.13 Lm^3 × 0.81 LF = 1.72 Bm^3

(2.4 yd^3 × 1.15 BFF = 2.76 LCY × 0.81 LF = 2.24 BCY)

Hourly Production —

1.72 Bm^3 × 75 cycles/h = 129 Bm^3/h

(2.24 BCY × 75 cycles/hr = 168 BCY/hr)

More accurate production estimates can be made by recording actual machine cycle times in the same or similar application. Then visually verify the approximate bucket fill factor.

Efficiency Considerations

Loader capacity should always be matched to peak production requirements of the job. Actual "on-the-job" loader productivity will be influenced by factors such as operator skill, personal delays, job layout and other delays. Experience and knowledge of local conditions will be the best indicators of actual job efficiency.

Operation	Working Hour	Efficiency Factor
Day	50 min/Hr	0.83

An Alternative Machine Selection Method

Another method of selecting the right Track Loader and bucket to meet production requirements is by use of the nomographs on the following pages. The method is quicker and easier than the proceding example because it does not require as many calculations, yet the accuracy is about the same within the normal limits of input data.

Be careful when entering and reading data from the nomographs because some scales increase from bottom to top, while others are the reverse. Do not be overly concerned with the precision as affected by pencil line width or reading to the hundredth of a m^3 (yd^3). Remember that bucket fill factor, material density, and cycle time are at best close estimates.

Example problem

A track loader must produce 200 Lm^3 (262 LCY) per hour. Estimated cycle time is 0.5 minutes, working 50 minutes per hour. Bucket fill factor is 110% and the material density is 1600 kg/Lm^3 (2700 lb/LCY).

Determine bucket size, machine model and hourly production in tons and yards.

Solution

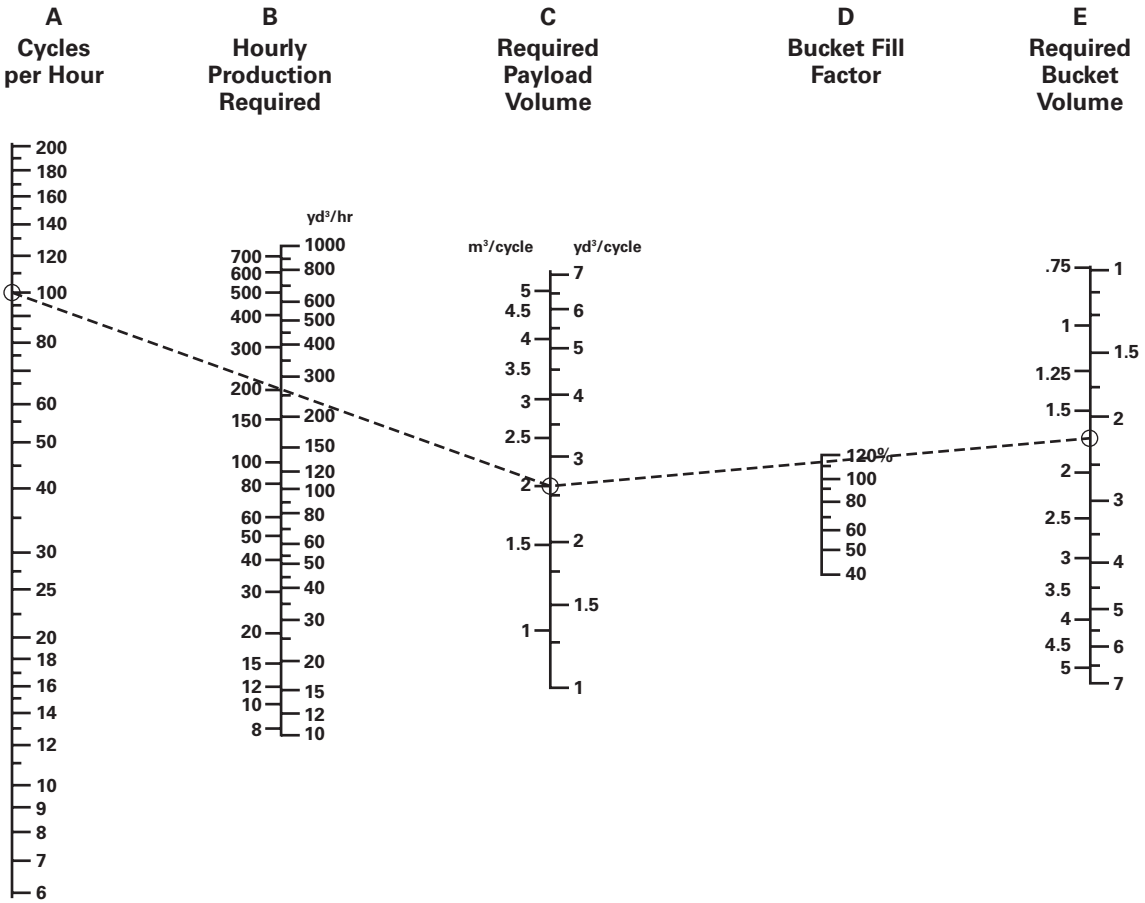
At full efficiency, it will cycle 120 times per hour. Since only an average 50 minutes are available, only 100 cycles will be completed per hour.

Starting on Scale A at 100 cycles per hour draw a straight line intersecting 200 m^3/hr (262 yd^3/hr) on Scale B and continuing the line on to Scale C giving 2.0 m^3 (2.62 yd^3) required payload.

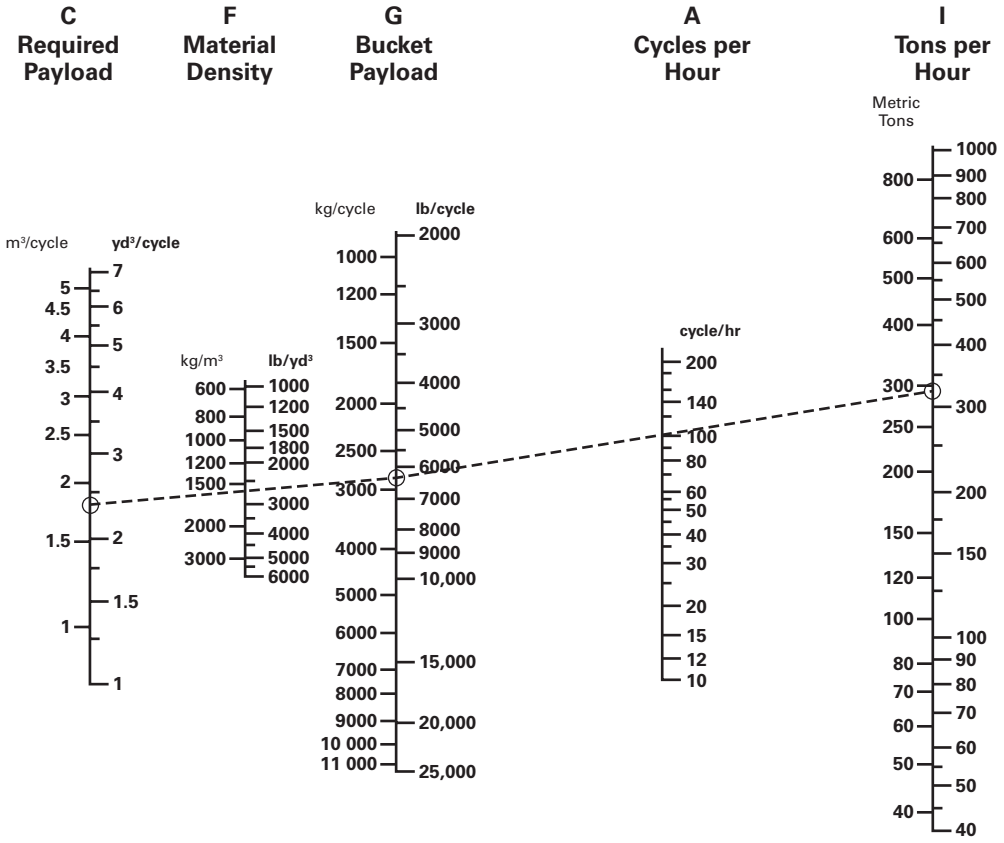
Follow steps 1 through 7 on the next two pages.

● ● ●

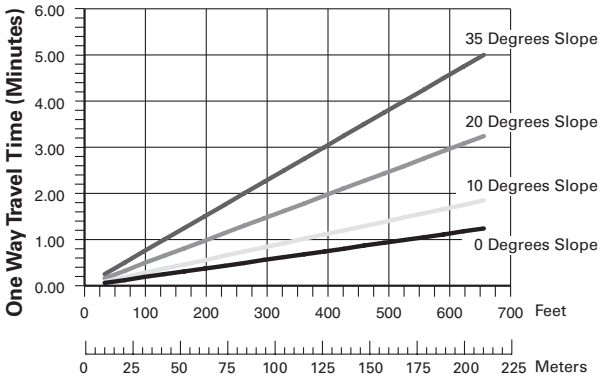
1) Enter Scale A cycles per hour (100) and B hourly production 200 m^3/hr (262 yd^3/hr).
2) Connect A and B and extend to C to find required payload 2.0 m^3 (2.62 yd^3).
3) Connect C to bucket fill factor on Scale D (110%) and extend to E to find required bucket size 1.8 m^3 (2.35 yd^3).
4) Transfer Scale A and C readings to nomograph on following page.



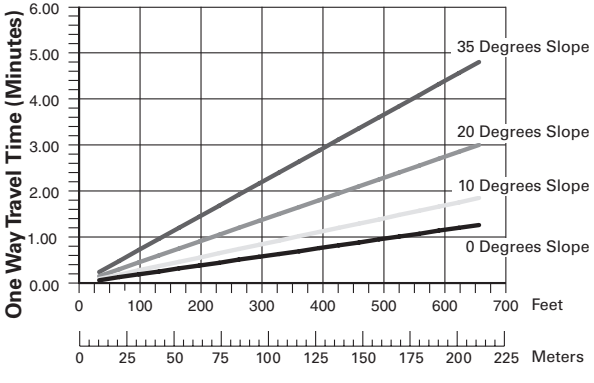
5) Connect C 1.8 m^3 (2.35 yd^3) to F 1600 kg/m^3 (2700 lb/yd^3) and extend to G to find payload weight 2880 kg (6345 lb).
6) Compare G bucket payload weight 2880 kg (6345 lb) with maximum operating capacities table in this section to see if the 1.85 m^3 (2.4 yd^3) bucket can handle the desired payload. Table indicates the 953D with a 1.85 m^3 (2.4 yd^3) bucket equipped with bolt-on cutting edge or teeth and segments has a greater operating capacity of 3343 kg (7370 lb), therefore stability is okay.
7) Extend Scale G reading 2880 kg (6345 lb) through Scale A (100) to Scale I to find tons per hour 288 metric ton/hr (317 U.S. ton/hr).



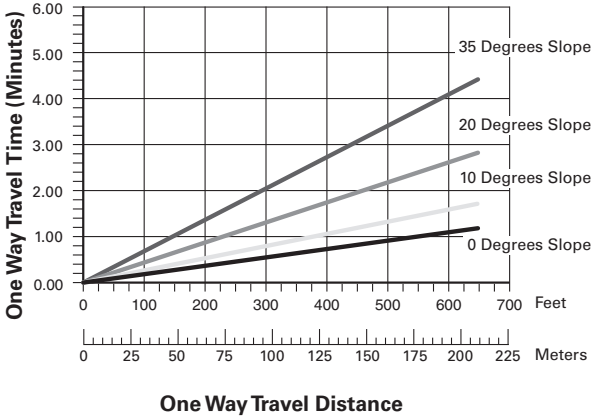
953D/953K



963D/963K



973D



TRAVEL TIME CHARTS

Conditions:

- No grades.
- Speeds loaded and empty essentially the same.
- Bucket position constant during travel.
- Travel encountered in maneuver portion of cycle not included.
- Acceleration time accounted for in maneuver time.

Travel Time (in minutes) =

$$Metric - \frac{\text{number of meters traveled}}{\text{speed (in km/h)} \times 16.67}$$

$$English - \frac{\text{number of feet traveled}}{\text{speed (in mph)} \times 88}$$

KEY

953D/953K — Hydrostatic top speed both forward and reverse 10 km/h **(6.2 mph)**
963D/963K — Hydrostatic top speed both forward and reverse 10 km/h **(6.2 mph)**
973D — Hydrostatic top speed both forward and reverse 11 km/h **(6.83 mph)**

Production Estimating Table
- m^3 or yd^3/60 min. hour
- Estimated bucket payload in bank m^3 or yd^3

Bucket Size (m^3 or yd^3)		1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Cycle Time Hundredths of a minute	**Cycles Per Hr**	**Unshaded area indicates average work range**								
0.25	240	240	360	480	600	720	840	960		
0.30	200	200	300	400	500	600	700	800		
0.35	171	171	257	342	428	513	599	684	769	
0.40	150	150	225	300	375	450	525	600	675	750
0.45	133	133	200	268	332	400	466	530	600	665
0.50	120	120	180	240	300	360	420	480	540	600
0.55	109	109	164	218	272	328	382	436	490	545
0.60	100	100	150	200	250	300	350	400	450	600
0.65	92	92	138	184	230	276	322	368	416	460

Work Tools	953D/953K	963D/963K	973D
Fusion™ Coupler	X (953K only)	X	
Quick Coupler	X	X	X
Fusion General Purpose Bucket	X (953K only)	X	
General Purpose (GP) bucket	X	X	X
GP bucket with trash rack	X	X	X
MP bucket with trash rack	X	X	X
Landfill bucket	X	X	X
Landfill Multi-purpose	X	X	X
Multi-purpose (MP) bucket	X	X	X
Trim blade	X	X	X
Slag bucket			X
Fusion Pallet Forks	X (953K only)	X	
Forks (for QC or bucket)	X	X	X
Material handling arm	X	X	X
Loader rake	X	X	X
Top-Clamp bucket	X		X
Side Dump bucket	X	X	X

Fusion Coupler System for Track Loaders

This coupler system is a performance multiplier for Track Loaders, giving owners many times the value over a machine with only a pin-on bucket.

The biggest gain for the track loader user is durability; the Fusion coupler endures like no other. Dual wedge locks pull the tool tight to the coupler and hold it there for a solid, rattle-free fit. No rattle means no wear, so tools last longer. The wedge lock is built to stay tight even after thousands of tool changes. This coupler is built to be as tough as the linkage itself, for trouble-free performance day in and day out.



The Track Loaders Fusion Coupler is supported by a set of tools:



General Purpose Buckets provide good all-around performance for stockpiling, re-handling, excavating and bank loading. A wide bucket is available for use with a machine that has wide gauge tracks.



Multi-Purpose Buckets have a unique four-way action that can load, strip top soil, bulldoze, clamp pipe or large chunks of concrete, clean up debris and many other tasks.



Pallet Forks are an essential tool on construction job sites. Cat Fusion Pallet Forks can be configured to allow the tines to either swing or stay in a fixed position.

Fusion Interchangeability

The Fusion Coupler for the track loaders is compatible with most of the Fusion tools for wheel loaders, allowing interchangeability. However, common sense must be used as machine power envelopes and use profiles are very different for each machine platform.

Care must be taken to get the best performance while taking advantage of tool interchangeability. While all Fusion couplers can pick up any Fusion tool, some tools are less useful when used on a different type of carrier than their design intent.

For compatibility questions, contact Cat Work Tools.

SHOE OPTIONS



① Double Grouser Shoes



② Trapezoidal Center Hole Shoes



③ Single Grouser Shoes



④ Chopper Shoes

- **Extreme service shoes** are available which have more hardened wear material for longer wear life and higher impact applications.
- **Wider shoes** are also available to reduce ground pressure in soft underfoot conditions.

Other shoe options are available. Consult a Cat dealer for more information.

SYSTEMONE SHOES



Double Grouser Shoes

- Work best in applications that require less penetration and traction.
- Recommended for applications that require better turning capability and less ground disturbance.
- Feature two or three short grousers instead of one tall grouser.



Center Hole Shoes

- Work best in applications where packing causes the track to tighten. They are recommended for applications with large amounts of debris what tend to pack in the track.
- Reduces extricable packing between the shoe and the bushing since they allow the sprocket to punch out dirt and debris.

Other shoe options are available. Consult a Cat dealer for more information.

TRACK-TYPE TRACTORS
Hydraulic Controls
Bulldozers
Rippers

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TRACK-TYPE TRACTORS

Features:

- **Cat® Diesel Engines** provide the power, high torque rise, reliability and performance you can depend on.
- **HEUI™** on D6R and D7R increases fuel efficiency, reduces smoke, improves cold starting and enhances diagnostic capabilities.
- **Mechanical Electronic Unit Injector (MEUI™)** on D8T, D9, D10 and D11 excels in its ability to control injection pressure over the entire engine operating speed range. It combines the technical advancement of an electronic control system with the simplicity of direct mechanically controlled unit fuel injection. These features allow the engine to have complete control over injection timing, duration, and pressure.
- **Common Rail fuel injection system** on D1, D2, D3, D6K2, D5R2, D6N, D6T, D6, D6XE and D7E machines; optimizes performance and fuel consumption, minimizes heat rejection, and lowers emissions.

- **Oil cooled steering clutches and brakes** standard on D6GC, D9, D10 and D11.
- **Finger Tip Controls (FTC)** of transmission, steering clutches and brakes on D10T2 and D11.
- **Differential steering** allows infinitely variable turning radius. Standard on the D5R2, D6N, D6R2, D6T, D7R, D7E, D8R, D8T, D9, D5, D6, D6 XE, D7 and D8 GC, allows the tractor to make a "power turn" keeping both tracks working for more traction and higher performance.
- **Electronic Hydrostatic Power Train System** on D1 through D4 allows power turns, stepless speed range, smooth modulation, dynamic hydrostatic braking, superior maneuverability and excellent controllability.
- **Electric Drive Power Train System** on D6 XE and D7E allows stepless speed range, smooth modulation, and excellent efficiency. When coupled with differential steer it provides superior maneuverability with locked-track pivot turn capability and excellent controllability.
- **Combined hand lever steering** located left of operator provides easier operation on D9R.
- **Standard Tractors** designed for heavy dozing and general grading.
- **XL Tractors D6T, D5R2 and D6R2** offer higher horsepower and longer roller frames for increased finish grading capability, flotation and productivity.
- **Extra Wide (XW) gauge** on D6T length roller frame provides wider shoes for greater flotation and stability for steep slope grading.
- **Sealed and Lubricated Track** reduces pin and bushing wear for lower undercarriage repair costs.
- **Cat Abrasion Undercarriage** extends undercarriage system life, improves reliability, and reduces owning and operating costs.
- **Elevated sprockets** (not on D4 or D7E) eliminate final drive stress induced by roller frame movement and ground impact loads. Final drives pull chain only. Seals moved up out of dirt, sand and water for longer life. Blade visibility improved because operator sits higher.
- **Resilient mounted bogie undercarriage** on D8R, D8T, D9, D10 and D11 reduces shock transmitted to tractor. Allows track to conform to rough ground for better traction.
- **Solid mounted undercarriage** standard on D1 through D3 provides stable platform for low impact, and high abrasion applications. Provides optimum finish grading performance.
- **Oscillating undercarriage** on D6K2 through D7E and optional on the D8R and D8T decreases ground shock to the machine and provides a smoother, more comfortable ride for the operator.
- **Accessible modular design** on D5, D6N XL and up greatly reduces drive train removal and installation time resulting in reduced repair costs.
- **Tag link** on D7, D7R, D8R/D8T and up; L-shaped push arms on D5, D6, D6R2, D6N, D6T and D7E. Both designs allow closer mounting of dozer blades. This reduces total tractor length, improves maneuverability, balance, blade penetration and pryout.
- **Low ground pressure (LGP)** tractors offer greater flotation in soft, swampy conditions. Available on D1 through D8T.

EXTREME SLOPE OPERATION

The maximum fore and aft grade in static condition on which each track-type tractor or pipelayer will maintain proper lubrication is 45 degrees (100%). Consult Operation & Maintenance Manual (if applicable) for POWERTRAIN fluid level overfill requirements for operation on extreme slopes. Extreme slope operation is anytime the slope exceeds 35°.

The ENGINE should never be overfilled with oil. This may lead to rapid overheating. For extreme slope operation, engine oil should be maintained at the full mark.

NOTE: Both ENGINE and POWERTRAIN fluid levels should be checked on level ground before working sidehills and slopes.

When working sidehills and slopes, consideration should be given to the following important points:

- Speed of travel — At higher speeds, inertia forces tend to make the tractor less stable.
- Roughness of terrain or surface — Ample allowance should be made where the terrain or surface is uneven.
- Mounted equipment — Bulldozers, sidebooms, winches, and other mounted equipment cause the tractor to balance differently.
- Nature of surface — New earthen fills may give way with the weight of the tractor. Rocky surfaces may promote side slipping of tractor.
- Track slippage due to excessive loads — This may cause downhill track to "dig in," increasing angle of tractor.
- Implements hitched to the drawbar — This may decrease weight on uphill track, e.g., logging arch, two-wheel wagon.
- Height of hitch on tractor — When a high drawbar is used the tractor is less stable than with the standard drawbar.
- Width of shoes — Wide track shoes tend to decrease "digging in", hence tractor is more stable.
- Operated equipment — Be aware of the stability and other performance features of the equipment operated by the tractor.
- Keep all attachments or pulled loads low to the ground for optimum stability.

NOTE: Safe operation on steep slopes may require special machine maintenance as well as excellent operator skill and proper equipment for the specific application. Consult Operation & Maintenance Manual (if applicable) for proper fluid level requirements.

Features:

- **Designed and built for specific tractor applications.** Valves and components sized for exacting quality and performance.
- **Job requirements matched** through various arrangements.
- **Hydraulic blade and ripper controls:** Mechanical controls on G Series. Electro hydraulic controls on D6N and D6K2. Pilot blade and ripper controls on D6T Tier 3/Stage IIIA, Japan 2006 (Tier 3) equivalent with optional electro hydraulic blade control. Electro hydraulic blade and ripper controls on D6T Tier 4 Interim/ Stage IIIB/Japan 2011 (Tier 4 Interim) equivalent Mechanical controls on D9R. Electro hydraulic blade and ripper controls on D7E, D8T, D9, D10 and D11.
- **Full flow filters***… all oil completely filtered.
- **Dual tilt** — standard on D11 and D11 CD, attachment option on D7E, D8R, D8T, D9 and D10.

*Exception — D8R 2-pump.

BULLDOZERS

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Features:

- **Straight Bulldozers** — adjustable tilt angle controls blade penetration.
- **Variable cutting edge Power Angle and Tilt (VPAT)** — blade available on certain models. The blade can be mechanically tipped forward for improved penetration or back for more productivity and easier finish grading.
- **Angling Bulldozers** — 25° right/left angling; C-frame allows mounting other tools.
- **Universal Bulldozers** — 25° wings provide increased capacity, less spillage.
- **Semi-Universal Bulldozers** — combines penetration ability of straight blade with increased load capacity provided by short 25° wings.
- **Wheel Dozer blades** are offered in straight and universal blade design with hydraulic pitch and tilt control.
- **Box-section construction** on blades adds rigidity and strength.
- **Cutting edges** are heat treated and reversible for extra life.
- **Performance Cutting edge** available for D6, D7 and D8.

BLADE SELECTION

Properly matching tractor and dozer is a basic requirement for maximizing production. First consider the kind of work the tractor will be doing most of its life. Then evaluate:

- Material to be moved.
- Tractor limitations.

Materials to be moved

Most materials are dozeable. However, dozer performance will vary with material characteristics such as:

Particle Size and Shape — The larger the individual particle size, the harder it is for a cutting edge to penetrate. Particles with sharp edges resist the natural rolling action of a dozer blade. These particles require more horsepower to move than a similar volume of material with rounded edges.

Voids — Few voids or the absence of voids means the individual particles have most or all of their surface area in contact with other particles. This forms a bond which must be broken. A well graded material, which lacks voids, is generally heavy, and will be hard to remove from the bank state.

Water Content — In most materials the lack of moisture increases the bond between particles and makes the material difficult to remove from the bank state. A high moisture content makes dozing difficult because the material is heavy and requires more force to move. Optimum moisture reduces dust and offers the best condition for dozing ease and operator comfort.

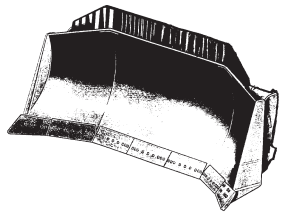
The effect of freezing depends on the moisture content. When frozen, the material's bond strengthens as moisture content increases and temperature decreases. However, freezing a completely dry material does not change its characteristics.

An indication of a blade's ability to penetrate and obtain a blade load is kW per meter (or horsepower per foot) of cutting edge. The higher the kW/meter (HP/foot), the more aggressive the blade. Kilowatt per Lm^3 (horsepower per loose cubic yard) indicates a blade's ability to push material. The higher the kW/Lm^3 (HP/LCY), the greater the blade's potential capability for carrying material at a greater speed.

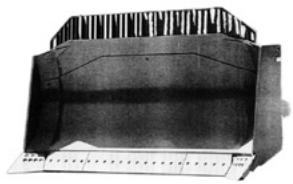
Tractor Limitations

The weight and horsepower of the machine determines its ability to push. No tractor can exert more pounds push than the machine itself weighs and its power train can develop. Various terrain and underfoot conditions on the job limit the tractor's ability to use its weight and horsepower. The "approximate coefficient of traction factors" chart in the Tables Section presents these traction factors for common materials. To use the chart, take the total tractor weight (with attachments) times the factor to arrive at the maximum usable push the dozer can exert.

Production Dozing Tools



"U" — Universal blade — the large wings on this blade include one end bit and at least one section of cutting edge which make it efficient for moving big loads over long distances as in land reclamation, stockpile work, charging hoppers and trapping for loaders. As this blade has a lower kW/meter (HP/foot) of cutting edge than an "S" or "SU", penetration should not be a prime objective. With a lower kW/Lm^3 (HP/LCY) than an "S" or "SU", this blade is best for lighter or relatively easily dozed material. If equipped with tilt cylinders the U blade can be used to pry out, level, cut ditches and steer the tractor.



"SU" — The Semi-U blade combines the desirable characteristics of S and U-blades into one package. It has increased capacity by the addition of short wings which include only the dozer end bits. The wings provide improved load retention capabilities while maintaining the blade's ability to penetrate and load quickly in tightly packed materials and to handle a wide variety of materials in production oriented applications. Tilt cylinder(s) increase both the productivity and versatility of this dozer. Equipped with a push plate, it is effectively used for push loading scrapers.